

SHARQAWY WAEI

Cairo, Egypt · sherqo@sherqo.me · sherqo.me · +20-109-121-5660
linkedin.com/in/sharqawycs · github.com/sherqo · leetcode.com/u/sharqawycs

SUMMARY

Computer & AI Engineering student focused on backend development, computer networks and systems-level engineering.

EDUCATION

B.Sc. Computer & Artificial Intelligence Engineering (Dual Degree) 2023 – 2027
Faculty of Engineering, Ain Shams University — Cairo, Egypt GPA: 3.4/4.0

B.Sc. Computer Science — University of East London, UK First Class Honours

EXPERIENCE

Full Stack Software Engineer Intern Jul 2026 – Aug 2026
Siemens Digital Industry Software / Siemens EDA Cairo, Egypt

- Built a visualization library rendering interactive diagrams at **50k+ node** scale for hardware verification pipelines
- Designed a real-time sync protocol between a **C++** backend and **React** frontend
- Used **Web Workers** to keep the UI responsive during heavy layout and validation operations
- Built bottleneck detection on live performance metrics

PROJECTS

Diran — Full-Stack Developer [Code] [Website]

- Built a Notion-style **collaborative block editor** with **real-time multi-user editing** via WebSockets, AI content suggestions, team management, role-based permissions, and full-text block search using **Next.js**, **Fastify.js**, and **PostgreSQL**
- Engineered complete auth system: **JWT** access/refresh tokens, **OTP** email verification, password reset, and rate limiting

Distributed LLM Serving System [Code]

- Built a distributed system in **Golang** and **Python** handling **1000+ concurrent** LLM inference requests with a custom **load balancer**, agent-based auto-scaling across multiple machines, **gRPC-based** fault detection, and zero request loss on node failure

Packet Pulse Protocol — Computer Networks Protocol [Code]

- Designed and implemented a custom **UDP-based application-layer protocol** in **Python** for IoT telemetry from scratch, featuring a **12-byte binary header**, message batching, sequence tracking, ACK handling, and loss-tolerant delivery
- Developed reproducible network experiment infrastructure including **PCAP** capture tooling, network emulation setup, and performance visualization scripts

SKILLS

Languages: C/C++, Java, JavaScript/TypeScript, Python, Go, Assembly, Shell
Frameworks: Node.js, Express.js, Fastify.js, React, Next.js, Astro, Svelte
Databases: PostgreSQL, MySQL, MongoDB, SQL Server
Tools: Git/GitHub, Docker, Linux, AWS (S3, EC2, Lambda), GitHub Actions, Vercel, Cloudflare Workers
Core Knowledge: OOP, OS Internals, Data Structures & Algorithms, Networking, Embedded Systems

EXTRACURRICULARS

- Solved **500+ problems** on **LeetCode** and completed **competitive programming training** with **Coach Academy** with exceptional performance [Certificate]